

Computational Fluid Dynamics - 14/6/2023

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We want to simulate the natural convection thermal plume generated by a hot cylinder immersed in a colder fluid. The flow is governed by the time-dependent incompressible Navier-Stokes equations with the *Boussinesq buoyancy model*, namely:

$$\begin{cases} \frac{\partial \mathbf{u}}{\partial t} - \nu \Delta \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p + \beta \mathbf{g}(T - T_{\text{ref}}) = 0, & \text{in } \Omega, t > 0, \\ \text{div } \mathbf{u} = 0, & \text{in } \Omega, t > 0, \\ \frac{\partial T}{\partial t} - \alpha \Delta T + \mathbf{u} \cdot \nabla T = 0, & \text{in } \Omega, t > 0, \end{cases} \quad (1)$$

where $\nu = 0.01$ is the kinematic viscosity, $\beta = 0.0002$ is the thermal expansion coefficient, $\alpha = 0.001$ is the thermal diffusivity, $\mathbf{g} = (0, -9.81)$ is the gravity acceleration and $T_{\text{ref}} = 20$ is a reference temperature.

On velocity and pressure, we consider the following boundary conditions:

- no-slip condition ($\mathbf{u} = \mathbf{0}$) on the bottom and lateral boundaries as well as on the cylinder;
- homogeneous Neumann condition ($(-p\mathbf{I} + \nu \nabla \mathbf{u})\mathbf{n} = \mathbf{0}$) on the top boundary;

while on temperature we impose

- the reference temperature ($T = 20$) on the bottom and lateral walls and a higher temperature ($T = 60$) on the cylinder;
- homogeneous Neumann condition ($\nabla T \cdot \mathbf{n} = 0$) on the top boundary.

The flow is initially at rest ($\mathbf{u} = \mathbf{0}, p = 0$ at $t = 0$) at the reference temperature ($T = T_{\text{ref}}$ at $t = 0$).

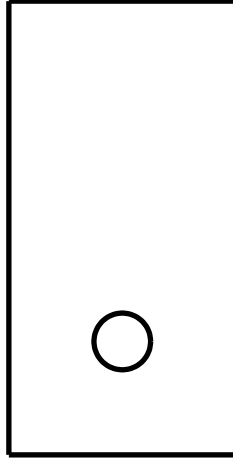


Figure 1: Computational domain

a) Introduce a suitable splitting-based time scheme for problem (1), such that

- the continuity and momentum equation are discretized by the incremental Chorin-Temam method;
- the temperature equation is discretized by the implicit Euler scheme;

- suitable choices are made on the time discretization of the coupling terms, to uncouple the Navier-Stokes equations from the temperature equation;
 - each subproblem is linear.
- b) Write the weak formulation of each sub-step discussing how the boundary conditions associated to each sub-step are imposed.
- c) Based on the template `exam.edp`, write a `FreeFEM++` script for the solution of the problem considering P_2 finite elements for velocity, P_1 for pressure and P_2 for temperature.
- d) Solve the problem using a time step $\Delta t = 1$ until the final time $T_f = 200$. Compute and display the time evolution of the average temperature on the top boundary. Moreover, compute and report the forces acting on the cylinder at the final time.